

Day 6 Evidence Workbooklet

Branch Routing with if and else

Engineering practice → condition result → branch followed → selected action → support route

Student copy. Guided workbooklet, not the mastery quiz. Print format: duplex, left-stapled packet.

Name		Section		Date	
Score	____/42		Mastery check	secure / developing / needs support	

The Week 2 scenario starts with a sensor-kit checkout table. A kit may be released, charged, sent to calibration, or held for missing checkout information. Today the focus is a single branch decision: read one condition, choose the branch that runs, and explain the action selected.

Engineering task	Decide which checkout action a sensor kit should receive from one condition at a time.
Computational move	Turn a true/false condition into one selected route while keeping the evidence for that route visible.
Python focus	<code>if</code> , <code>else</code> , condition result, selected branch, action variable, and printed route evidence.
Evidence to keep	case value, condition expression, <code>True/False</code> result, branch followed, action assigned, and support note.

Day 6 working rules

1. Python checks the condition after `if`.
2. If the condition is `True`, the `if` branch runs.
3. If the condition is `False`, the `else` branch runs.
4. Only one of those two branches runs.
5. A reliable branch note names both the condition result and the selected action.

1 Read the checkout condition before choosing an action

recognize

predict

The lab team is checking whether a sensor kit can leave the checkout table. Day 6 uses one decision at a time: if the condition is true, one action happens; otherwise the other action happens.

```
1 kit_label = "S-14"
2 charge_percent = 82
3 charge_minimum = 80
4 calibration_label = "current"
5 borrower_initials = "JL"
6
7 charge_ok = charge_percent >= charge_minimum
8 calibration_ok = calibration_label == "current"
9 borrower_logged = borrower_initials != ""
10
11 if charge_ok:
12     action = "send to calibration check"
13 else:
14     action = "charge before checkout"
```

Pts.	Prompt	My evidence
1	Which variable stores the kit label?	
1	Which comparison creates charge_ok?	
1	Is charge_ok true or false for this kit? Show the numbers.	
1	Which branch assigns the action for this kit?	

2 Trace an if/else route

[trace](#) [explain](#)

Complete the route table. Use the condition result to decide which branch runs.

Pts.	Charge value	Condition	Branch followed	Action and reason
2	82	<code>charge_percent</code> <code>>= 80</code>		
2	80	<code>charge_percent</code> <code>>= 80</code>		
2	79	<code>charge_percent</code> <code>>= 80</code>		
2	Explain why the exact value 80 matters.			

Boundary connection

The route depends on a comparison from Week 1. If the comparison is wrong, the branch route will be wrong too.

3 Choose the selected action from printed evidence

[predict](#) [explain](#)

The same structure is used for calibration status. Read the code and predict the action.

```
1 calibration_label = "expired"
2 calibration_ok = calibration_label == "current"
3
4 if calibration_ok:
5     checkout_status = "ready for borrower log"
6 else:
7     checkout_status = "send to calibration bench"
8
9 print(kit_label, checkout_status)
```

Pts.	Prompt	My evidence
2	What value is stored in <code>calibration_ok</code> ? Why?	
2	Which branch runs: the <code>if</code> branch or the <code>else</code> branch?	
2	What exact status text is printed after the kit label?	
2	Why would <code>bool(calibration_label)</code> be unsafe here?	

4 Complete one branch condition

[implement](#) [debug](#)

Complete the condition so the kit moves to borrower logging only when the borrower initials are present.

```
if _____:
    next_step = "record borrower and release kit"
else:
    next_step = "hold until borrower initials are entered"
```

Pts.	Prompt	My response
3	Write the completed condition using <code>borrower_initials</code> .	
2	Why should the condition compare against an empty string instead of checking the kit label?	
2	Give one input value that should follow the <code>else</code> branch.	
1	Name the evidence type you used: state / type / boundary / branch.	

5 Debug a branch-routing claim

[debug](#) [test](#) [explain](#)

A teammate writes: "The kit is ready because `charge_ok` is true." Use branch-routing evidence to evaluate that claim.

Pts.	Prompt	My response
2	What part of the claim is supported by the code?	
2	What evidence is still missing before a full checkout release?	
2	Give one test case that should follow the hold/repair route.	
2	Write a safer one-sentence checkout note.	

Pts.	My checkout-routing note
4	Write a short note that says how an if/else route chooses one action from one condition. Use the sensor-kit checkout example.
2	Circle the support route you would use next: condition result / branch followed / exact boundary / string label / written explanation. Name one next action: _____

Bridge to Day 7

Day 6 chooses between two actions. Day 7 adds ordered rules where the first true condition wins and later branches are skipped.